SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Divya Dharshini S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Divya

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	2	2	2	2
Solution Process	25%	2	2	2	3	
Accuracy of Calculation	20%	2	2	2	2	
Interpretation of Results	15%	2	1	1	1	

Total Marks 38 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWER

Q1. Subnetting 192.168.20.0/24 into Lab A (/26), Lab B (/27), Lab C (/28)

Given: Address block 192.168.20.0/24 (mask 255.255.255.0, 256 addresses in total). Lab A = /26, Lab B = /27, Lab C = /28.

Method: Block size = 256 − (last octet of the mask). Usable hosts = 2^h − 2, where h is the number of host bits. Subnets are allocated largest first, one after another, so they don't overlap.

Step 1: Convert each prefix length into a mask, host bits and block size

Lab	Prefix	Last octet of mask (binary = decimal)	Host bits (h)	Block size	Total addresses	Usable hosts
A	/26	11000000 = 192	32 − 26 = 6	256 − 192 = 64	$2^6 = 64$	$2^6 - 2 = 62$
B	/27	11100000 = 224	32 − 27 = 5	256 − 224 = 32	$2^5 = 32$	$2^5 - 2 = 30$
C	/28	11110000 = 240	32 − 28 = 4	256 − 240 = 16	$2^4 = 16$	$2^4 - 2 = 14$

Step 2: Allocate the subnets one after another, largest first

- Lab A starts at 192.168.20.0 and, with 64 addresses, ends at .63.
- Lab B starts at the next free address, .64 (a multiple of 32, so it is a valid /27 boundary), and ends at .95.
- Lab C starts at .96 (a multiple of 16, so it is a valid /28 boundary), and ends at .111.

Step 3: Apply the rules for each subnet

- Network address = first address of the block.
- Broadcast address = last address of the block (all host bits set to 1) = next network address − 1.
- Valid host range = network address + 1 up to broadcast address − 1.
- Usable hosts = 2^h − 2, because the network and broadcast addresses cannot be assigned to devices.

Result

Lab	Mask	Block size	Network	Valid host range	Broadcast	Usable hosts
A	/26 (255.255.255.192)	64	192.168.20.0	.1 – .62	192.168.20.63	$2^6 - 2 = 62$
B	/27 (255.255.255.224)	32	192.168.20.64	.65 – .94	192.168.20.95	$2^5 - 2 = 30$
C	/28 (255.255.255.240)	16	192.168.20.96	.97 – .110	192.168.20.111	$2^4 - 2 = 14$

Verification

- Addresses used: 64 + 32 + 16 = 112, which is exactly .0 to .111, so there is no gap and no overlap between the labs.
- Addresses left in the /24: 256 − 112 = 144, i.e. 192.168.20.112 – 192.168.20.255. This can later be split into, for example, 192.168.20.112/28 (16 addresses) and 192.168.20.128/25 (128 addresses).
- Total usable hosts across the three labs: 62 + 30 + 14 = 106.

Interpretation: The three labs use 112 addresses, from .0 to .111. The range 192.168.20.112 – .255 is still free for future growth. Lab A can host up to 62 devices, Lab B 30 and Lab C 14, so each lab should be sized to its expected number of computers plus some spare capacity.

Q2. Systems 192.168.1.130 and 192.168.1.190, mask 255.255.255.192

Given: System 1 = 192.168.1.130 and System 2 = 192.168.1.190, both with subnet mask 255.255.255.192.

Method: 255.255.255.192 is /26, so the block size is 64 and the subnet boundaries are 0, 64, 128 and 192. Network address = the IP ANDed with the mask.

Step 1: Identify the prefix, host bits and block size

- 255.255.255.192 = 11111111.11111111.11111111.11000000, which has 8 + 8 + 8 + 2 = 26 network bits, so the prefix is /26.
- Host bits = 32 − 26 = 6, so each subnet has $2^6 = 64$ addresses and the block size is 256 − 192 = 64.
- The /26 subnets of 192.168.1.0 are: 0–63, 64–127, 128–191 and 192–255 (in the last octet).

Step 2: AND each IP address with the mask (last octet)

- System 1: 130 = 10000010; mask 192 = 11000000; 130 AND 192 = 10000000 = 128. 130 lies in 128–191.
- System 2: 190 = 10111110; mask 192 = 11000000; 190 AND 192 = 10000000 = 128. 190 lies in 128–191.

Step 3: Find the broadcast address and host range

- Broadcast = network address + block size − 1 = 128 + 64 − 1 = 191 (host bits all 1s: 10111111 = 191).
- First host = 128 + 1 = 129 and last host = 191 − 1 = 190.

Result

	System 1	System 2
IP address	192.168.1.130	192.168.1.190
Subnet range	192.168.1.128 – 192.168.1.191	192.168.1.128 – 192.168.1.191
Network address	192.168.1.128	192.168.1.128
Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
Broadcast address	192.168.1.191	192.168.1.191
Within its subnet range?	Yes	Yes (last usable host)
Usable hosts	$2^6 - 2 = 62$	62

Verification

- System 1: 129 ≤ 130 ≤ 190, so 192.168.1.130 is a valid host address.
- System 2: 129 ≤ 190 ≤ 190, so 192.168.1.190 is a valid host address (the last one), and it is neither the network address (.128) nor the broadcast address (.191).

Interpretation: Both systems are in the same subnet (192.168.1.128/26). They can communicate directly without a router. 192.168.1.190 is the last assignable address, so only .191 (the broadcast) comes after it. The subnet can hold 62 hosts, of which 2 are already in use here, leaving 60 free addresses.

Q3. Subnetting 172.16.0.0/16 for HR (/25), Sales (/26), IT (/27), Admin (/28)

Given: Address block 172.16.0.0/16 (65,536 addresses). HR = /25, Sales = /26, IT = /27, Admin = /28.

Method: Allocate in descending order of size, starting from 172.16.0.0, with each subnet beginning where the previous one ended.

Step 1: Convert each prefix length into a mask, host bits and block size

Dept	Prefix	Mask	Host bits (h)	Block size	Total addresses	Usable hosts
HR	/25	255.255.255.128	7	256 − 128 = 128	$2^7 = 128$	$2^7 - 2 = 126$
Sales	/26	255.255.255.192	6	256 − 192 = 64	$2^6 = 64$	$2^6 - 2 = 62$
IT	/27	255.255.255.224	5	256 − 224 = 32	$2^5 = 32$	$2^5 - 2 = 30$
Admin	/28	255.255.255.240	4	256 − 240 = 16	$2^4 = 16$	$2^4 - 2 = 14$

Because every prefix is longer than /24, all four subnets fit inside the single range 172.16.0.0 – 172.16.0.255 and only the last octet changes.

Step 2: Allocate the subnets in descending order of size

- HR (128 addresses): 172.16.0.0 to 172.16.0.127.
- Sales (64 addresses): starts at .128 (a multiple of 64) and ends at .191.
- IT (32 addresses): starts at .192 (a multiple of 32) and ends at .223.
- Admin (16 addresses): starts at .224 (a multiple of 16) and ends at .239.

Result

Dept	Mask	Block size	Network	Valid host range	Broadcast	Usable hosts
HR	/25	128	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	$2^7 - 2 = 126$
Sales	/26	64	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	$2^6 - 2 = 62$
IT	/27	32	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	$2^5 - 2 = 30$
Admin	/28	16	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	$2^4 - 2 = 14$

Unused range: The departments use 128 + 64 + 32 + 16 = 240 addresses (172.16.0.0 – 172.16.0.239). The unused block is 172.16.0.240 – 172.16.255.255, which is 65,536 − 240 = 65,296 addresses.

Verification

- The unused space is made up of 172.16.0.240 – 172.16.0.255 (16 addresses) plus 172.16.1.0 – 172.16.255.255 (255 × 256 = 65,280 addresses): 16 + 65,280 = 65,296.
- Total usable hosts assigned: 126 + 62 + 30 + 14 = 232.
- Share of the /16 in use: 240 ÷ 65,536 ≈ 0.37%.

Interpretation: Allocating the largest subnet first avoids fragmentation and wasted space. Almost the entire /16 remains available for future subnets. HR can hold 126 devices, Sales 62, IT 30 and Admin 14.

Q4. 172.16.10.0/24 with a /27 mask

Given: Network 172.16.10.0/24 subnetted with a /27 mask (255.255.255.224). Host to locate: 172.16.10.78. Address to check: 172.16.10.95.

Method

- Subnet bits borrowed = 27 − 24 = 3, so the number of subnets = 2^3 = 8.
- Host bits = 32 − 27 = 5, so usable hosts per subnet = 2^5 − 2 = 30.
- Block size = 256 − 224 = 32. The subnets start at 0, 32, 64, 96, 128, 160, 192 and 224.

Step 1: List all eight subnets

Subnet	Network address	Valid host range	Broadcast address
1	172.16.10.0	172.16.10.1 – 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 – 172.16.10.62	172.16.10.63
3	172.16.10.64	172.16.10.65 – 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 – 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 – 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 – 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 – 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 – 172.16.10.254	172.16.10.255

Step 2: Find the subnet of 172.16.10.78

Subnet for 172.16.10.78: 78 falls between 64 and 95, so it belongs to the third subnet. Check by ANDing with the mask: 78 = 01001110; mask 224 = 11100000; 78 AND 224 = 01000000 = 64.

- Network address: 172.16.10.64
- Valid host range: 172.16.10.65 – 172.16.10.94
- Broadcast address: 172.16.10.95 (64 + 32 − 1)

Step 3: Check 172.16.10.95

Is 172.16.10.95 in the same subnet? It is inside the subnet range (64–95), but it is the broadcast address. It belongs to this subnet but cannot be assigned to a host. Check: 95 = 01011111; 95 AND 224 = 01000000 = 64, the same network address as 172.16.10.78. Its five host bits are all 1s, which confirms that it is the broadcast address.

Verification

- Total addresses: 8 subnets × 32 = 256, which matches the /24.
- Total usable hosts: 8 × 30 = 240 (16 addresses are lost to the 8 network and 8 broadcast addresses).

Interpretation: 172.16.10.78 is a valid host in subnet 172.16.10.64/27, and 172.16.10.95 is that subnet's broadcast address. The /27 mask creates 8 subnets of 30 usable hosts each, which suits departments with up to 30 devices.

Q5. IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329

Given: An IPv6 address is 128 bits long, written as 8 groups of 16 bits (4 hexadecimal digits each), separated by colons. The given address is 2001:0db8:0000:0000:0000:ff00:0042:8329 and the prefix length is /64.

Compression (shortest form):

1. Drop leading zeros in each group: 2001:db8:0:0:0:ff00:42:8329
2. Replace the longest run of all-zero groups with :: (allowed only once): 2001:db8::ff00:42:8329

Here the longest run of consecutive zero groups is groups 3, 4 and 5 (three groups), so those are replaced by ::. The :: can be used only once in an address, because using it twice would make the number of missing groups ambiguous.

Expansion back to full form:

1. Count the groups that are written: 2001, db8, ff00, 42 and 8329 = 5 groups. An address needs 8 groups, so :: stands for 8 − 5 = 3 missing zero groups: 2001:db8:0:0:0:ff00:42:8329
2. Pad each group to 4 hex digits: 2001:0db8:0000:0000:0000:ff00:0042:8329

This matches the original address, so the compression is correct.

Network prefix and interface ID

Network prefix (/64): The first 64 bits are the first 4 groups (64 ÷ 16 = 4): 2001:0db8:0000:0000. The prefix is 2001:db8::/64 (2001:db8:0:0::/64). The remaining 64 bits (0000:ff00:0042:8329, or 0:ff00:42:8329) form the interface ID that identifies the host.

Number of host addresses: Host bits = 128 − 64 = 64, so the total is 2^{64} = 18,446,744,073,709,551,616 addresses (about 1.8×10^{19}). IPv6 has no broadcast address, so essentially all of these are usable. The all-zeros interface ID is conventionally reserved as the subnet-router anycast address.

Interpretation: A /64 is the standard IPv6 LAN subnet size. Its address space is so large that it never limits the number of hosts. The prefix 2001:db8::/32 is reserved for documentation examples, so this address is a sample address and not one that is routed on the real Internet.

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